INAI 6986 · AI Capstone · Milestone 2 Workstream

Milestone 2 — Model Testing, Validation & Refinement

Current Development Report

Development Results — Final Evaluation Pending

This report documents completed development-phase testing, the corrected scoring analysis, the iteration narrative, and the honest state of the project. The protected final benchmark campaign has not been run: the final-system freeze and final-run approval remain open human decisions. This document is current-stage complete; it is not equivalent to a completed final Milestone 2 evaluation.

Student	J.O.
Course	INAI 6986 — AI Capstone (Milestone 2 workstream)
Report Status	Development-phase report — personally reviewed and edited against the Milestone 2 rubric on 2026-10-08; final-evaluation gates remain open; no full-credit score claim
Date Prepared	October 8, 2026 (afternoon, CDT) — day-phase dating, not backdated
Evidence Boundary	Parent-frozen class evidence snapshot, 2026-10-08 13:42 Central Daylight Time
Evaluation Authority	Human reviewer retains final authority; machine advisory scores are labeled as advisory throughout

Identity-protected public copy prepared from the reviewed private grading document.

1. Summary of Results

Plain statement of where the project stands: On a frozen set of 24 non-protected development questions, four locally fine-tuned model families (each at a 1,600-example training size) were run in two conditions — the fine-tuned adapter and its matched, untouched base model — for 192 evaluation slots. Under the corrected, conservative definition (an answer counts only when it is answered, factually correct, supported by a valid citation, and carries no unsupported claims), the proposed candidate — the Nemotron N1600 adapter — achieved 4 of 24 successful answers, identical to its matched base (4 of 24). Gemma and Ornith adapters improved over their own bases (0 → 4 and 1 → 4 respectively) on this advisory measure. No family met the business thresholds carried from the project plan (a 10-percentage-point adapter improvement over the matched base, at least 95 percent valid evidence locators, under 2 percent unsupported claims, and 90 percent correct abstention behavior). None of these numbers is a final accuracy claim: the scoring is a post-hoc local machine advisory pass, not human-adjudicated accuracy, and the protected final test set has not been opened. In business terms: the system is not ready to support eligibility or policy decisions, and no autonomous eligibility determination is claimed or intended — a human reviewer remains the decision authority.

Bottom line: development-phase testing is complete and honestly reported; the threshold was not met on advisory development metrics; the final gated evaluation is pending explicit human approval.

2. Experimental Setup

Pipeline and frozen configuration. All four candidate families were instruction-tuned adapters trained locally at matched data size (1,600 training examples per family, selected from token-fit-gated size ladders of 200/509/750/1,600; cross-tokenizer training losses were never ranked against one another). Each adapter was evaluated side-by-side with its matched untouched base under identical serving settings on one frozen, question-only development set.

Development Question Set	24 frozen questions (question-only JSONL; development split; non-protected). 3 questions per source document across 9 source documents.
Conditions	4 families × 2 conditions (matched base, fine-tuned adapter) × 24 slots = 192 slots
Scoring Protocol	advisory_judge_v2 — seven-dimension post-hoc development rescoring; primary = Answered AND Correct AND Supported AND No-Unsupported-Claims, over all 24 slots per condition
Judge	Local Qwen worker model (qwen3.8-flash-next-nvfp4) on an approved local endpoint; no external API authored or scored anything
Verification	Every 192-slot replay verified against source result hashes; original v1 journals and generations unchanged; zero judge retries in the Oct 8 v1/v2 passes
Human Review Completed	Batch B submitted (revision 11): 30 review cards over 15 paired questions (includes preservation probes); 6 dimensions = 180 ratings. This is error-selected human review — NOT 30 independently verified errors, and not population accuracy.

Test Protection	The protected final benchmark, its answer keys, and the canonical protected files were never opened during development; no development artifact reads them

Change from the project plan and reason: the original single-score success protocol was withdrawn after it inflated a refusal-heavy condition (reported 24/24 through 23 abstentions). The corrective, pre-declared conservative definition above replaced it, and v1 outputs were preserved unmodified for audit. A later user-accepted scope decision (2026-10-08, recorded verbatim) set the final benchmark to fresh official sources with AI-generated, source-verified question authorship — explicitly not student-authored — with the original 30 completed student-authored controls preserved and no new student-written questions required.

3. Iteration Narrative

From baseline selection to the current proposed candidate, drawn from run records:

- Training-stage gates: all 16 fine-tuning runs passed token-fit gating across four families and four data sizes. Because tokenizers differ across families, next-token loss was recorded for fit diagnostics only and never used to rank factual quality.
- Withdrawn first scoring protocol (V1): the original protocol scored an abstention-heavy condition artificially high (Ornith base 24/24, of which 23 were refusals to answer). V1 was withdrawn as a success metric — its correct ranking lesson is preserved — while every V1 journal, binding, and generation remains untouched and replay-verified.
- Corrected rescore (V2, 2026-10-08): a seven-dimension advisory rescore over the same unchanged outputs. 150 of 192 slots produced accepted judgments; 19 were budget failures (generation could not complete) and 23 were adapter-parser failures (Muse's native conversation template produced responses the structured-answer parser could not read). Zero judge requests were unresolved and no judge retries occurred.
- Targeted refinement experiment (2026-10-07, Nemotron lineage only): 8 paired questions re-run, producing 16 new responses from the same selected Nemotron adapter (checkpoint 400). Outcome disclosed honestly: the refinement added failed and blanket-base-abstention responses, and 8 invalid adapter locators remained unchanged. The refinement was not promoted and is not claimed as a successful model fix.
- Candidate proposal (not selection): Nemotron N1600 with matched base was proposed by the student and accepted as the direction of work on 2026-10-08. Rationale: it ties the best observed development score (4/24), completed all 48 generation slots, and shares lineage with the submitted human error review. It is explicitly not a proven factual winner — Gemma and Ornith adapters tie it, and small observed differences with unknown judge failures do not establish a statistical winner.

Stage	What Was Tried	What Happened	Conclusion
Token-fit gating (16 runs)	4 families × sizes 200–1,600	All 16 runs completed fit gates	Size ladder informative within family only; no cross-family loss ranking
V1 success scoring	Single-score protocol on 192 slots	Refusal-inflated 24/24 exposed	Withdrawn as metric; artifacts preserved; corrective lesson recorded
V2 advisory rescore	7 dimensions, conservative conjunction	150 judgments; 17 successes; 48 abstentions	Bases and adapters compared on same footing; no final claim
Oct 7 refinement (8 pairs)	Re-run paired failures, Nemotron adapter	Added failures/abstentions;	Negative finding; not promoted

		locators unchanged	
Human Batch B (rev 11)	30 review cards over 15 paired questions (incl. preservation probes), 6 dimensions	180 ratings submitted	Human error-selected review exists; card count must not be read as 30 verified errors

4. Test Results — Final Evaluation Pending

Explicit boundary, kept visible so this section is never mistaken for the final run: the protected final test set has NOT been opened and the one-time final run is not authorized. The rubric's Test Results heading is retained deliberately; the tables below are development-phase results under the corrected advisory metric, presented in the same comparison structure (matched baseline vs. model, same metric, same split) so the final gated run will slot directly beside them when approved. A confusion-matrix-equivalent and business-unit translation for the final model are withheld as pending, not fabricated.

4.1 Answer Success By Condition (Successful Answers / 24 Slots)

Model Family (N1600)	Matched Base	Fine-Tuned Adapter	Generation Failures	Judge/Parser Failures
Nemotron N1600	4/24	4/24	0 base / 0 adapter	0 total non-judged
Gemma N1600	0/24	4/24	0 base / 1 adapter	1 total non-judged
Ornith N1600	1/24	4/24	0 base / 1 adapter	1 total non-judged
Muse N1600	0/24	0/24	16 base / 24 adapter	23 parser failures (adapter); 16 terminal budget failures (base)

Across all conditions: 192 slots, 150 accepted judgments, 19 budget failures, 23 parser failures, 17 successful answers, 48 abstentions. These aggregates are descriptive development observations, not a deployment accuracy claim.

4.2 Abstention Behavior

Abstention Dimension (150 Judgments)	Count
Not Applicable	88
Failed To Abstain	35
Unnecessary	11
Appropriate	16

4.3 Threshold Scorecard (Development Advisory Values; Not Final)

Objective Carried From Plan	Development Observation (Advisory)	Final Status
≥10-point adapter improvement over matched base	Nemotron +0 (4/24 vs 4/24); Gemma and Ornith +4/24 each on this measure	Not met on advisory development data; final test pending
≥95% valid evidence locators	Adapter conditions show 14–17 invalid locators of 23–24 judged slots per family	Not met; final test pending
<2% unsupported claims	18–19 unsupported-claim-present judgments of 23–24 in each	Not met; final test pending

	Nemotron/Gemma/Ornith adapter (Nemotron base also 17; Ornith base 0 — its errors were abstentions)	
90% correct abstention behavior	35 failed-to-abstain and 11 unnecessary-abstain judgments of 150	Not met; final test pending

5. Error And Robustness Analysis

Named failure modes with counts, derived from the 150 accepted advisory judgments across all eight conditions (categories carried forward from the Week 8 error taxonomy and re-checked):

Error Category	Count / 150	What It Suggests
Wrong Value	57	Data change: retrieval or passage-selection misses the governing figure/date; strengthen evidence selection
Abstention Issue	32	Scope rule: explicit abstention policy and calibrated refusal training needed (both over- and under-abstention observed)
Hallucinated Rule	26	Modeling change: rule-statement grounding; penalize ungrounded normative claims in fine-tuning targets
No Error	26	Correct-and-supported answers; the reference class for adjudication
Invalid Locator	8	Data/format change: citation targets that do not resolve; locator validation gate before any answer is served
Missing Evidence	1	Pipeline change: required evidence absent from the prompt window

Robustness: What Was Actually Tested

- Source-sensitivity / refinement stress (completed, negative finding): re-running 8 paired failure questions with the selected Nemotron adapter produced 16 new responses whose net effect was negative — additional failures and a blanket base-condition abstention pattern, with 8 invalid adapter locators unchanged. This is disclosed as evidence that naive iteration does not repair the failure modes.
- Format robustness (incidental, disclosed): the Muse family exposed a real robustness failure class — 23 of 24 adapter responses were unparseable through the structured-answer parser because of the model's native response template, plus 16 base-side terminal budget failures. Its 0/24 is a harness-compatibility result, not 24 incorrect-policy judgments.
- Not yet run (final-gated): the rubric's deployment-change robustness check (missing/malformed fields, later time period, rephrased or truncated text), seed stability, and any time-savings comparison. No robustness-movement number is claimed for the frozen final candidate because that run is pending approval.

6. Subgroup Performance

Performance broken out by source-document subgroup (the segments of the policy corpus the decision touches), derived mechanically from the 192 recorded outcomes joined to the frozen question-only set — no semantic re-grading of chat content. Successes are counted out of 8 slots per document (4 families × 2 conditions).

Source Document ID	Questions	Slots	Successful Answers	Answered	Abstained
0b35d55bccccd7f2	3	24	1	12	7
0cf7f74661264c1c	3	24	4	14	6
0fedf0e92a0512d5	1	8	0	3	4
40da7233f00682c0	3	24	3	14	4
4610161564ff40a6	3	24	3	14	6
6041319ef53009c3	3	24	0	10	7
61d87699f5a94b77	3	24	1	14	4
7ab755bb5012a435	3	24	4	14	5
945ab0be1933d150	2	16	1	7	5

Gaps as found: success is concentrated in a minority of source documents — two documents (60413…, 0fedf…) produced zero successful answers in every condition, while two others (0cf7f…, 7ab75…) account for 8 of the 17 successes. Answered-but-unsupported judgments dominate every document's answered slots (9–13 of 10–14 per document), indicating per-document evidence-selection weakness rather than a uniform model deficit. Planned response: per-document failure characterization feeds the fresh-source benchmark design; subgroup gates will be re-checked on the final run. Governance note: under NIST AI Risk Management Framework terms this is a documented Identify/Manage cycle (disparate document-level reliability disclosed, human review mandated); an eligibility or benefits decision made on these models without a human would raise the kind of high-risk-concern that EU AI Act high-risk obligations address — which is precisely why autonomy is out of scope here.

7. Limitations And Readiness

- Advisory, not final: every score here was produced by a local machine judge under a post-hoc corrected protocol. It is not human-adjudicated accuracy, and the pre-declared final protocol will run once, after freeze, on a fresh protected benchmark that has not been opened.
- Small denominators: 24 slots per condition; 150 of 192 slots produced judgments. The three adapters tied at 4/24 — no statistical winner exists in this data.
- Judge pipeline honesty: no judge retries in the October 8 passes; the separate October 7 refinement advisory had 5 initial schema failures disclosed separately. 48 abstentions and 17 positive judgments are descriptive aggregates.
- Preserved corrective history: the withdrawn V1 protocol and its refusal-inflation lesson are preserved verbatim; V1 outputs are unchanged. The 19 originally exposed diagnostics remain exposed — no 'untouched' claim is made about them.

- Human dispositions already completed (Human Batch B, revision 11, error-selected) are not re-opened or double-counted; prior document inventories predating the October 7–8 parent proofs are superseded by this report's verified sources.
- Instrument validity is an open gap, not a closed one: the Milestone 1 feedback correctly noted that the original 30 controls test document navigation rather than policy reasoning, contained no unanswerable items, and concentrated on a few pages. Those originals stay preserved but are historical-test-exposed and were not re-scored. Nine student-adopted unsupported-case items (AI-assisted drafts, disclosed as such) and the fresh-source benchmark target exactly this abstention-coverage gap; until they are frozen and run, the 90% correct-abstention objective has no adequate denominator, and the current 24-question development set does not exercise exceptions, effective dates, conflicts, or cross-document synthesis. The advisory numbers above therefore characterize supported-answer behavior on lookup-type questions only.
- Course standing: instructor permission for the LLM-based student-reported workflow was confirmed by the student; no documented waiver of rubric criteria and no guaranteed grade is claimed. Business decisions about eligibility or policy remain human authority; the system makes no autonomous determination.

Readiness: Current-Stage Complete Vs. Final Gated

Gate	State
Development question set frozen and 192-slot campaign replay-verified	Complete
Corrected v2 advisory rescore verified against source hashes	Complete
Human error review (Batch B rev 11: 30 cards / 15 paired questions, 180 dimension ratings)	Submitted
Targeted refinement experiment executed and honestly negative (Nemotron lineage)	Complete
Proposed candidate accepted (Nemotron N1600 + matched base; fresh-source benchmark scope)	User-accepted proposal — not a system freeze
Fresh benchmark source/exposure/key verification	Open
Immutable final-system and key freeze with explicit human approval	Open
One-time final and declared-robustness run on protected benchmark	Not authorized
Final human adjudication and final report QA	Open
Reproduction notebook executed offline over raw artifacts	Complete 2026-10-08 — executed .ipynb/HTML/PDF; 192/192 slots recomputed

8. Reproducibility And AI Use Disclosure

8.1 Evidence Ledger (Read-Only Sources Of Record)

Evidence	Path (Absolute)	SHA-256 (First 16)
Parent-frozen class evidence snapshot	<REPO_ROOT>/artifacts/milestone2_completion/class_package_20261008/evidence_snapshot.json	230836193433202a
Verified advisory v2 results	<REPO_ROOT>/artifacts/milestone2_completion/development/	e1a6ceaa8479d91c

	shortlist_v1/verification/parent_advisory_v2_actual_results_verified_20261008.json	
Verified gate reconciliation	<REPO_ROOT>/artifacts/milestone2_completion/development/shortlist_v1/verification/parent_development_gate_reconciliation_verified_20261008.json	4b49a39a3e716cfa
User scope decision (verbatim)	<REPO_ROOT>/artifacts/milestone2_completion/development/shortlist_v1/verification/user_candidate_benchmark_scope_decision_20261008.json	6bcf2098b031cf32
Per-slot counted outcomes (192 rows)	<REPO_ROOT>/artifacts/milestone2_completion/development/shortlist_v1/verification/development_v2_counted_outcomes_20261008.jsonl	38ebb4e21e329106
Frozen question set (24, question-only)	<REPO_ROOT>/artifacts/milestone2_completion/development/shortlist_v1/frozen/questions.jsonl	c9185bb14e9f437b

This builder recomputes every displayed figure directly from the counted-outcomes file and the frozen question set; displayed totals were asserted against the parent snapshot at build time. Candidate lineage: NVIDIA-Nemotron-3.5-Lightning-30B-A3B-BF16, revision a9904d24bcc1d289a1950fa9d2b978c47cf903b9, adapter SHA-256 4aaca5c564ed518ddbc8cce87f887a61998fd9c4039d4abd7e00185c4dc0a65a, checkpoint-400. A repository link is not embedded because no externally published repository URL has been verified as safe for disclosure; that is an open pre-submission item, not a gap concealed as complete.

8.2 AI Use Disclosure (Required Disclosure)

AI assistance: Hermes agent management and approved local worker models (local Qwen judge for advisory scoring; local endpoints for all fine-tuning and generation) executed documented, supervised tasks. No external chat service authored candidate outputs or benchmark questions; no external API trained anything. Human authorship and accountability: the student conceived the project, chose architecture, data, models, and evaluation decisions, performed hardware/environment setup and troubleshooting, directed the agents, made every proceed/stop/retry/quarantine choice, corrected factual errors found in this report's drafts during final review, re-ran the reproducibility notebook end to end, and checked every reported number against its evidence file, and accepts full responsibility for all claims. Machine scores herein are advisory drafts and this document carries no final approval or full-credit grade claim.

8.3 Reproduction Note

A companion reproducibility notebook (reproducibility/Development_Reproducibility_Replay_20261008.ipynb) was executed end-to-end on 2026-10-08: it recomputes the hash chains, raw-response bindings, per-condition numerators, and all 192 counted slot records from raw artifacts, rebuilds the scrubbed portable bundle (reproducibility/data/, 28 files, HASHES.json re-hashed clean), and cross-checks the sealed release manifest. It performs 0 model reruns and 0 network calls; executed .ipynb, HTML and PDF are committed alongside the replay module and its unit test.

Appendix A — Extended Human-Contributed Work And Tooling Record

Supplemental detail only. This appendix is separate from, and does not replace, the required AI Use Disclosure in Section 8.2.

Student-attributed work: project conception and business framing; architecture and family selection; the original 30 independent control questions (student-authored, completed, preserved); direction and troubleshooting of hardware and environment failures; the recorded proceed/stop/retry/quarantine decisions (withdrawing V1, not promoting the negative refinement, accepting the Nemotron proposal and the fresh-source benchmark scope on 2026-10-08); and final review and acceptance of every number in this report. The development questions were produced by the approved local pipeline with source verification and are not claimed as student-authored; the fresh-benchmark authorship-method change is disclosed and is not an instructor waiver.

Tooling record: Hermes coordinated agent management; all training and generation ran on local hardware via approved local endpoints; the advisory judge was the local Qwen worker (qwen3.8-flash-next-nvfp4). No measured human-hours figure exists, and none is stated. Public identity: J.O..